

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

UC MATH 600 Mathematics for Computer Science — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Lehman, Leighton & Meyer — Mathematics for Computer Science (free book)	The primary text, free in full. This course follows its development closely, and its exercises are the natural problem set. It is written for exactly this audience.
MIT 6.042J / 6.1200J Mathematics for Computer Science (OCW, full video)	The lectures, free. Watch them alongside the modules — the recitation videos are where the proof technique is actually demonstrated.
Velleman — How to Prove It	Modules 01 through 03. The best book on proof technique as a skill rather than a subject, and it is unusually patient. Library copy.
Rosen — Discrete Mathematics and Its Applications	Modules 04 through 08 as a reference. Comprehensive and dry; good for looking something up, poor for reading straight through. Library copy.
Mitzenmacher & Upfal — Probability and Computing	Modules 11 and 12. Where discrete probability is developed for computer scientists rather than for statisticians, and the direct prerequisite for CSCE 658. Library copy.
Cormen, Leiserson, Rivest & Stein, chapters 3 and 4	Modules 09 and 10. Asymptotics and recurrences in the form CSCE 629 will use them. Library copy.

Module-by-module map

Module 01 — What a Proof Is For

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapter 1 (free)	§§1 and 2 — what a proof is, with the false proofs that make the point better than true ones do.
Velleman — How to Prove It, chapter 1	§1 — proof as a transferable skill, patiently. Library copy.
Pólya — How to Solve It	§3 — how to approach a problem you cannot yet do, which is the actual difficulty. Library copy.
Project Euler (free)	§2's workflow — problems where testing a conjecture computationally is the natural first move.

Module 02 — Logic and Proof Technique

Resource	What it covers for this module
Velleman — How to Prove It, chapters 1 to 3	The whole module — and the quantifier chapter is worth more than its length suggests. Library copy.
Lehman, Leighton & Meyer, chapters 1 and 3 (free)	§§1 to 3, free — with propositional logic developed enough to make the negation rules obvious.
MIT 6.042J lectures 1 and 2 (free video)	§§2 and 3 — the techniques demonstrated on real statements rather than described.
Hammack — Book of Proof (free)	The whole module, free in full — and it has far more worked examples than this course can include.

Module 03 — Induction

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapter 5 (free)	The whole module , free — and the false induction proofs it works through are the best part.
MIT 6.042J, the induction lectures (free video)	§§1 to 3, with the horses example done properly.
Hammack — Book of Proof, chapter 10 (free)	§§1 and 3 — many worked examples, free, and strong induction is handled carefully.
CLRS, the loop invariant material (library copy)	§3's application — loop invariants are induction on iterations, and this is the form CSCE 629 will use from week one.

Module 04 — Sets, Functions, Relations

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapters 4 and 8 (free)	§§1 to 4, free — and the infinite-sets chapter is unusually careful.

Resource	What it covers for this module
Velleman, chapters 4 and 5	§§2 and 3 — relations and functions with the well-definedness obligation made explicit. Library copy.
Hammack — Book of Proof, chapters 11 and 14 (free)	§§3 and 4, free — relations, and cardinality with the diagonal argument worked slowly.
Sipser, chapter 4 (library copy)	§4's consequence — where the diagonal argument becomes undecidability, which is CSCE 627.

Module 05 — Number Theory and Modular Arithmetic

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapters 8 and 9 (free)	The whole module , free — and it develops RSA from this material, which is the right motivation.
MIT 6.042J, the number theory lectures (free video)	§§1 to 3, with Euclid proved rather than asserted.
Silverman — A Friendly Introduction to Number Theory	§§1 to 3 — patient, and the early chapters are available free from the author.
Project Euler, the early problems (free)	§§1 and 4 — problems where implementing Euclid and modular exponentiation is the natural solution.

Module 06 — Counting

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapter 15 (free)	The whole module , free — and the four rules are stated exactly as §1 gives them.
MIT 6.042J, the counting lectures (free video)	§§1 to 3, with the bijection rule used repeatedly.
Graham, Knuth & Patashnik — Concrete Mathematics	§§2 and 3 far beyond this course — demanding, excellent, and the binomial coefficient chapter is the reference. Library copy.
Hammack — Book of Proof, chapter 3 (free)	§§1 and 2, free, with many worked examples.

Module 07 — Graphs

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapters 11 to 13 (free)	The whole module , free — graphs, colouring, and matching developed together.

Resource	What it covers for this module
MIT 6.042J, the graph theory lectures (free video)	§§1 to 4, with the modelling emphasised.
West — Introduction to Graph Theory	§§2 and 4 in depth — the standard reference when you need a theorem. Library copy.
CLRS, the graph algorithm chapters (library copy)	§3's 'what to do next' column — which is CSCE 629's content, previewed.

Module 08 — Trees and Connectivity

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapter 12 (free)	§§1 to 3, free — trees and the equivalence proofs, done carefully.
MIT 6.042J, the trees lecture (free video)	§§1 and 2 — with the leaf argument of §2 made explicit.
CLRS, the tree and heap chapters (library copy)	§4 — where these counting facts become data structure bounds, which is CSCE 629.
Sedgewick & Wayne, Algorithms (free site and lectures)	§4 implemented — free, with code, and a good way to see the bounds hold empirically.

Module 09 — Asymptotics

Resource	What it covers for this module
CLRS, chapter 3 (library copy)	§§1 and 2 — the definitions in the form CSCE 629 uses, with the set-membership point made explicitly.
Lehman, Leighton & Meyer, chapter 14 (free)	§§1 to 3, free — asymptotics with the common abuses catalogued.
Knuth — Big Omicron and big Omega and big Theta (free)	§1's callout — two pages on the notation and why the equals sign is what it is, by the person who standardised it.
Sedgewick & Wayne, the analysis lectures (free)	§4 — measurement alongside analysis, which is the practice this module argues for.

Module 10 — Recurrences

Resource	What it covers for this module
CLRS, chapter 4 (library copy)	The whole module — the three methods, the master theorem's proof, and the substitution trap of §4 worked.
MIT 6.006, the divide-and-conquer lectures (free)	§§2 and 3 — recursion trees drawn, which is the right way to meet them.

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapter 21 (free)	§§2 and 4, free — recurrences solved by expansion and by guess-and-verify.
Mitzenmacher & Upfal, the recurrence material	§4 for randomised algorithms — where the recurrences become expectations, which is CSCE 658. Library copy.

Module 11 — Discrete Probability

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapters 17 to 19 (free)	The whole module , free — and the sample-space discipline of §1 is emphasised throughout, correctly.
Mitzenmacher & Upfal, chapters 1 to 3	§§1 to 3 — probability for computer scientists, and the direct prerequisite for CSCE 658. Library copy.
Gigerenzer — Reckoning with Risk	§3 — the frequency framing and why it works, with the medical examples. Library copy.
Tijms — Understanding Probability	§§1 and 4 — the paradoxes resolved by naming the sample space, which is the right way to meet them. Library copy.

Module 12 — Expectation and Concentration

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapters 19 and 20 (free)	The whole module , free — and the indicator variable examples are the best collection available.
Mitzenmacher & Upfal, chapters 2 to 4	§§1 to 4 — expectation, variance, and the Chernoff bounds proved. The direct bridge to CSCE 658. Library copy.
MIT 6.042J, the expectation and deviation lectures (free)	§§1 to 3, with linearity's independence-freedom emphasised properly.
Motwani & Raghavan — Randomized Algorithms	§3 applied — the bounds used in anger, which is CSCE 658's content. Library copy.

Module 13 — Claiming You Proved Something

Resource	What it covers for this module
Velleman — How to Prove It, the final chapters	§§1 and 2 — writing proofs for a reader, which is the skill this module is about. Library copy.
Lamport — How to Write a 21st Century Proof (free)	§2 — a structured proof format designed so that gaps are visible, by somebody who found errors in published work.

Resource	What it covers for this module
Lehman, Leighton & Meyer, the false proofs (free)	§2 — worked examples of arguments that look right and are not, which is the best practice material there is.
Pólya — How to Solve It, the looking-back chapter	§2 — the step after finding a proof, which is where the learning is. Library copy.